T3000 API & Protocol

Tstat LCD System — Documentation Index · built Nov 13, 2026

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This folder contains the complete technical documentation for the Tstat10/Tstat11 LCD UI system: a visual drag-and-drop screen designer that outputs JSON, rendered identically on browser, device, and phone.

Design Approach

Start with a blank 320×480 canvas. Drag widgets from a toolbox (label, text, input, dropdown, icon, divider, etc.) onto the canvas. Snap to the 17×10 character grid. Adjust position, font, alignment, and data bindings. Export as JSON. Sync to device.

Documents

Document	Description
System Overview	End-to-end architecture, data flow, and integration points
LCD Screen Designer	**Core document** — blank canvas, widget toolbox, grid snap, properties panel, drag-and-drop design flow
Widget Types	LVGL-style widget type reference (label, text, input, dropdown, icon, etc.)
JSON Schema	The `screenDefinition.json` format specification
Implementation Status	What's built, what's pending, build order

Key Specs

- **LCD Resolution:** 320 × 480 px (fixed)
- **Character Grid:** 17 columns × 10 rows
- **Font:** Bold monospace (Fira Mono / Consolas)
- **Background:** Temco Blue (#003366)
- **Devices:** Tstat10 (white unit), Tstat11 (32M dev board)
- **Transport:** JSON → Modbus register writes (no firmware flash)

Related Code

- Simulator: `src/t3-react/features/tstat10-simulator/`
- HVAC Draw Tool: `src/lib/t3-hvac/`
- Original prototype: `D:\1025\github\temcocontrols\Tstat10_Simulator\`